

Guneev Pannu

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EDUCATION

University of Toronto – 3.96 GPA

Sept 2024 – May 2028

HBSc Double Major in Computer Science and Statistics, Math Minor

Relevant Courses: Data Structures & Algorithms, Object -Oriented Development, Systems Programming, Database

EXPERIENCE

Course Software Developer

May 2026 – Present

University of Toronto

- Architected a live queuing and logging system for office hours, securing a Fall rollout with 500+ students
- Automated student enrollment, saving 2+ hrs of manual entry for professors by building a CSV pipeline
- Developed a Cron based email reminder service for interested students, increasing office hour attendance by 15%
- Integrated UofT SAML authentication for up to 15K+ students, eliminating manual account creation
- Engineered a scheduling dashboard to automate the creation of 50+ sessions/semester with recurring rules

Computer Science Education Research Assistant

May 2026 – Present

University of Toronto

- Authored weekly lab materials to deepen understanding of 500+ students for CSC108 Intro to Programming course
- Prepared weekly practice problems, removed 33% of existing redundant ones and replaced with self written
- Evaluated the effectiveness of LLM AI tools on potential course assignments, leading to adjustments of 15% of code

Frontend Developer

Sept 2025 – Present

University of Toronto

- Shipped 11 web pages to publicize the talent of 400 + club members in Next.js
- Built flexible components that fetched content from database, reducing manual updates by 60%
- Developed a content management system through custom forms to allow the editing of pages by non -tech people
- Developed an interview dashboard system to streamline recruitment for 40+ applicants

PROJECTS

Blog Website Backend | *Spring Boot, Java, Maven, Git, PostgreSQL, Docker*

- Built a backend software application in a **week** that could support a blog post social media app, like Tumblr
- Integrated **Spring Data JPA** to simplify communication from the service layer to the persistence layer
- Implemented JWT-based authentication and role -based authorization using **Spring Security**
- Integrated development tools like **Lombok** and **MapStruct** to reduce code boiler plater speeding development.

Paint Program | *Java, JavaFX, Maven, Git*

- Built a paint program that allowed users to draw 180+ shapes and lines on a resizable canvas using JavaFX
- Applied the MVC design pattern reducing merge conflicts by 30% , allowing for more rapid development
- Engineered a save - load system of the current canvas using finite state machines to achieve load times of <100 ms

Custom Unix Shell | *C, Make, Git*

- Developed a custom **Unix** shell in **C** replicating core Bash functionality, including process creation, piping, and command parsing
- Implemented inter-process communication and socket-based client-server chat system over local networks
- Debugged and memory-optimized system using GDB and Valgrind, achieving zero memory leaks

Python Syntax Checker | *PyTorch, Git*

- Trained a machine learning model to detect simple Python syntax errors with **92.5%** test accuracy
- Created and labeled a **5,000 sample dataset** using Python's ast and tokenizer modules

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R

Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries: pandas, NumPy, Matplotlib